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A PROJECT REPORT
ON

“MULTILINGUAL IMAGE CAPTION GENERATOR USING DEEP LEARNING”

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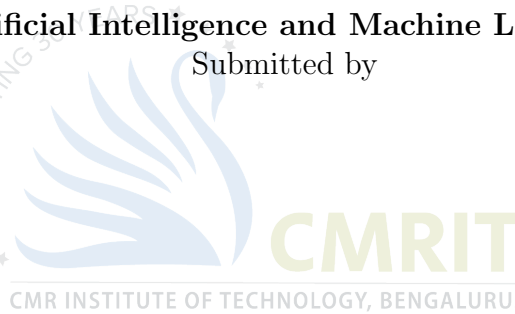
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Submitted by

MONIKA S
SAJANA R
SANIYA

1CR21AI035
1CR21AI048
1CR21AI052



Under the Guidance of

MS. ANSHIKA SHUKLA

Asst. Professor, Department of AIML, CMRIT



DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

CMR INSTITUTE OF TECHNOLOGY

AECS LAYOUT, ITPL PARK ROAD, BENGALURU - 560037

2024-2025

CMR Institute of Technology
AECS Layout, Bengaluru-560037
Department of Artificial Intelligence and Machine Learning



CERTIFICATE

Certified that the project work entitled **“MULTILINGUAL IMAGE
CAPTION GENERATOR USING DEEP LEARNING”** is carried out by
MONIKA S **1CR21AI035**
SAJANA R **1CR21AI048**
SANIYA **1CR21AI052**

a bonafide student of CMR Institute of Technology in partial fulfillment for the award of Bachelor of Engineering in Artificial Intelligence and Machine Learning of the Visvesvaraya Technological University, Belgaum during the year 2024-2025. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the department library.

The project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said Degree.



Ms. Anshika Shukla
Project Guide

Dr. Shyam P Joy
Head of the Department

Dr. Sanjay Jain
Principal

External Viva

Name of the examiners Signature with date

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- 2.

ABSTRACT

The rapid growth of digital media and the increasing demand for accessible content have driven the development of automated tools for generating textual descriptions of images. These systems combine computer vision and natural language processing to create meaningful captions, making visual content accessible to visually impaired users and improving content understanding across various applications. This project leverages ResNet50 for image feature extraction and LSTM for generating descriptive text, using the Flickr30K dataset to ensure adaptability to a wide range of image types and scenarios.

The system also incorporates multilingual output and audio capabilities, expanding its usability across languages and accessibility platforms. It has wide-ranging applications in accessibility for visually impaired individuals, media management for better indexing and searchability, and industries such as education, entertainment, security, and online content platforms. The model's performance is evaluated using the BLEU score, which measures the accuracy of the generated captions compared to reference captions, ensuring high-quality and contextually relevant output.

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MONIKA S
SAJANA R
SANIYA

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Chapter 1

INTRODUCTION

A Multilingual Image Caption Generator is a system designed to analyze visual content and produce meaningful textual captions in multiple languages that describe the scene, objects, actions, or context present in an image. This technology has gained prominence due to its wide-ranging applications in accessibility, content management, and human-computer interaction.

The primary objective of a multilingual image caption generator is to bridge the gap between visual and textual data while supporting diverse linguistic contexts. For instance, visually impaired individuals can benefit from descriptive captions in their native language, enabling them to perceive and understand images. Similarly, search engines and content recommendation systems can leverage image descriptions to improve indexing and retrieval accuracy. From social media platforms to e-commerce websites, the ability to automatically describe images in multiple languages can transform the way users interact with digital content globally.

In this project, we aim to design and implement a state-of-the-art multilingual image caption generator that leverages cutting-edge AI models such as ResNet50 and LSTM to produce high-quality captions in various languages, supported by the Google Text-to-Speech (gTTS) API for audio descriptions.

1.1 Challenges in Image Captioning

Developing a multilingual image caption generator presents several significant challenges across technical, linguistic, and contextual dimensions. One of the primary challenges lies in accurately interpreting the vast variability in image content, which includes diverse objects, scenes, and activities, as well as complex interactions between them. Images often contain ambiguous or context-dependent elements that are difficult for machines to interpret without deeper semantic understanding.

Another major challenge is generating captions that are not only factually correct but also linguistically fluent, coherent, and culturally sensitive across multiple languages. Ensuring that the translations are accurate and free of cultural biases is critical, especially when dealing with global datasets that reflect diverse perspectives.

Balancing conciseness and informativeness in captions becomes even more complex when adapting them to different linguistic structures.

Computational challenges also arise, as training and deploying deep learning models like ResNet50 and LSTM for multilingual captioning require substantial processing power and memory. Additionally, integrating multilingual capabilities demands robust translation APIs and mechanisms to manage artificial tokens for language-specific processing. Evaluating such systems is particularly difficult due to the subjective nature of language and the lack of universally accepted standards for multilingual captions. Overcoming these challenges necessitates innovative solutions in model architecture, multilingual dataset curation, and evaluation methodologies.

1.2 Motivation

The ability to describe images in natural language is a fundamental aspect of human cognition, enabling effective communication and understanding of the visual world. However, many individuals, particularly those with visual impairments, face significant barriers in accessing and comprehending visual content. This gap underscores the importance of developing a multilingual image caption generator to make images accessible to everyone, regardless of their physical abilities or linguistic preferences.

The growing volume of digital images shared daily across social media platforms, e-commerce websites, and other online domains further highlights the need for automated tools to interpret and describe visual data efficiently. Descriptive captions, particularly in multiple languages, can enhance search engine optimization (SEO), enabling more accurate indexing and retrieval of images, which is crucial for global businesses and users alike.

The rapid development of deep learning models, such as ResNet50, LSTM, and translation APIs, has provided a robust foundation for building sophisticated multilingual image description systems. These advancements inspire the pursuit of this project to explore how state-of-the-art techniques can be harnessed to tackle the inherent challenges of understanding and describing complex visual content across diverse languages. Ultimately, the motivation for this study stems from a desire to contribute to a more inclusive and connected digital environment, where artificial intelligence is leveraged to overcome barriers, enhance communication, and improve the quality of life for diverse populations.

1.3 Project Objectives

The primary objective of the Multilingual Image Caption Generator project is to design and implement an intelligent system capable of analyzing visual content and generating accurate, meaningful, and contextually appropriate textual descriptions in multiple languages. Specifically, the project seeks to:

- 1. Enhance Accessibility:** Develop a tool that generates image descriptions in various languages to aid visually impaired users in understanding and interacting with visual content.
- 2. Improve Content Understanding:** Enable better interpretation and organization of multimedia content for applications such as search engines, social media platforms, and content management systems.
- 3. Leverage Advanced AI Models:** Utilize cutting-edge architectures such as ResNet50 for feature extraction and LSTM for sequential caption generation.
- 4. Support Multilingual Capabilities:** Integrate artificial tokens and translation APIs to ensure robust support for multiple languages.
- 5. Ensure Linguistic and Contextual Quality:** Generate captions that are grammatically correct, contextually relevant, and culturally sensitive across languages.
- 6. Provide Audio Descriptions:** Use the gTTS API to produce audio descriptions of captions for improved accessibility.
- 7. Offer Quantifiable Evaluation:** Employ automated evaluation metrics like BLEU scores, combined with user feedback to assess and refine the system's performance.

1.4 Purpose and Significance

In today's digital era, images form a significant part of online communication, documentation, and entertainment. Social media platforms, e-commerce websites, educational resources, and accessibility tools heavily rely on visual content to engage users and convey information. However, the absence of textual descriptions for images in multiple languages creates accessibility challenges for visually impaired individuals, hindering their ability to comprehend visual data.

Furthermore, unstructured image data makes it difficult for search engines, content management systems, and AI-driven applications to process, retrieve, and organize images effectively. This growing reliance on visual content highlights the pressing need for a multilingual image caption generator, a system designed to bridge the gap between visual and textual data across languages.

The purpose of this study is to develop a robust and efficient system that leverages advanced artificial intelligence techniques to analyze images and generate accurate, meaningful, and contextually appropriate textual descriptions in multiple languages. This project aims to address key challenges such as the diversity of image content, linguistic coherence, and cultural sensitivity in generated captions. By integrating deep learning models like ResNet50 for visual feature extraction and LSTM for sequential text generation, coupled with translation APIs and gTTS for audio descriptions, the study seeks to push the boundaries of AI in understanding and describing visual content.

Chapter 2

LITERATURE REVIEW

Artificial intelligence (AI) and deep learning have become driving forces behind innovations across various domains, with image captioning standing out as one of the most impactful applications. As machines have become better at understanding visual content, there has been a growing emphasis on creating systems capable of generating human-like descriptions of images. This capability is essential not only for making visual information accessible to diverse audiences but also for advancing technologies in fields such as healthcare, autonomous vehicles, and digital media. The challenge, however, lies in creating models that not only generate descriptive captions but also understand the context, relationships between objects, and the linguistic nuances of different languages.

2.1 Introduction

Recent advancements in artificial intelligence (AI) and deep learning have profoundly impacted the field of image captioning, bringing it closer to achieving human-level performance. AI systems now have the ability to generate meaningful and contextually accurate textual descriptions from images, bridging the gap between visual content and natural language understanding. These systems typically employ sophisticated architectures such as Convolutional Neural Networks (CNNs), which excel at extracting visual features from images, and Recurrent Neural Networks (RNNs), which are adept at modeling sequential dependencies in language. The combination of these technologies has opened new possibilities for image captioning applications, enabling models to generate captions that are not only accurate but also contextually relevant.

In addition to CNNs and RNNs, several cutting-edge techniques have been integrated into image captioning models to enhance their performance. The incorporation of attention mechanisms, for instance, has allowed these models to focus on the most relevant parts of an image when generating captions, resulting in more accurate and nuanced descriptions. Furthermore, the development of multilingual captioning systems has enabled AI models to generate captions in multiple languages, broadening their applicability to global audiences. Transfer learning has also played a pivotal role,

allowing models to leverage pre-trained networks and adapt them to new tasks, significantly reducing the training time and computational costs. This chapter provides a comprehensive review of the key contributions in the domain of image captioning, discussing the evolution of methodologies, the challenges addressed by these innovations, and their transformative impact on the field.

2.2 Evolution of Image Captioning Systems

1.Template-Based Methods: Early approaches relied on predefined templates and object recognition, which were limited in flexibility and contextual understanding.

2.Deep Learning Revolution: The introduction of CNNs and RNNs transformed the field by allowing end-to-end learning of image-to-text relationships. Models like VGGNet and ResNet demonstrated superior performance in extracting deep visual features.

3.Attention Mechanisms: The incorporation of attention mechanisms significantly improved caption quality by enabling models to focus on relevant image regions during caption generation.

2.3 Key Techniques in Image Captioning

2.3.1 Encoder-Decoder Architectures

Encoder-decoder architectures are central to modern image captioning systems, as they define the framework through which images are converted into descriptive text. In these models, the encoder, typically a pre-trained deep convolutional neural network (CNN) like ResNet50, is responsible for extracting high-level features from an image. These features represent the image in a way that is useful for caption generation. The encoder transforms the image into a fixed-size vector or a set of feature maps that encapsulate key visual information, which is then passed on to the decoder. The decoder, often a Recurrent Neural Network (RNN) or Transformer, sequentially generates captions by interpreting the extracted features and mapping them into a coherent text sequence. The captioning process involves the generation of one word at a time, where each subsequent word is influenced by the previously generated words, ensuring that the caption is syntactically and semantically accurate.

The integration of attention mechanisms into this encoder-decoder framework has been a significant improvement in image captioning systems. Attention allows the model to dynamically focus on different parts of the image during the caption generation process, thereby enhancing the relevance and accuracy of the output. By weighting the importance of various features, the attention mechanism ensures that the model highlights the most pertinent aspects of the image when generating a partic-

ular word in the caption. This approach has improved interpretability, as researchers and practitioners can now visualize which parts of the image the model is attending to during each word generation. Attention mechanisms have thus played a critical role in boosting the performance of image captioning systems, leading to more precise and contextually aware captions.

2.3.2 Multilingual Image Captioning

Multilingual image captioning introduces significant challenges, primarily due to differences in grammar, sentence structure, and cultural context across languages. While generating captions in one language may be straightforward, extending this capability to multiple languages requires overcoming language-specific nuances. For example, languages like Japanese or German have different word orders, grammatical structures, and expressions, making it essential for image captioning systems to handle each language's characteristics independently. Furthermore, cultural context plays a crucial role in how certain objects or actions are described. A model must understand not only the linguistic features of a language but also the cultural subtleties to ensure that captions remain relevant and appropriate.

To address these challenges, Tsutsui and Crandall (2020) proposed the use of artificial tokens to manage multilingual caption generation within a single unified model. These tokens act as indicators that guide the model in selecting the language in which the caption should be generated. The inclusion of artificial tokens allows the model to seamlessly switch between languages while ensuring that each generated caption respects the grammatical and syntactic rules of the target language. This approach significantly improves the efficiency of multilingual captioning systems, as it eliminates the need for training separate models for each language. By using a single model, the system can provide high-quality, culturally and contextually relevant captions in multiple languages.

2.3.3 Transfer Learning

Transfer learning has become a cornerstone technique in modern image captioning systems, providing a means to leverage the knowledge contained in pre-trained models to accelerate the training process and enhance model performance. Transfer learning typically involves using a model that has been pre-trained on a large and diverse dataset, such as ImageNet, and adapting it to a more specific task, such as image captioning. This approach is particularly valuable because it allows the system to benefit from feature representations that have already been learned from a broader range of images, which can be directly applied to the task of caption generation.

By fine-tuning a pre-trained model like ResNet50 on a smaller, task-specific dataset, transfer learning reduces the need for vast amounts of labeled data and computational

resources. The system can now focus on learning the relationships between the extracted visual features and the language model, rather than starting from scratch in learning low-level image features. This approach also aids in improving the generalization of the model, allowing it to perform well on new, unseen data. In the context of image captioning, transfer learning enables models to generate more accurate and diverse captions in a shorter amount of time, making it an indispensable strategy in the development of efficient image captioning systems.

Feature	Traditional Models	Modern Approaches
Feature Extraction	VGG,AlexNet	ResNet50,ViT
Sequence Modeling	Simple RNNs	LSTMs,Transformers
Multilingual Support	Separate models	Artificial Tokens,Unified Models

Table 2.1: Comparison of Key Techniques

2.4 Related Studies

2.4.1 CNN-RNN Framework for Multi-Label Image Classification

Wang et al. (2016) introduced a novel unified model that combines Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) to address the challenge of multi-label image classification. CNNs are well-known for their ability to extract spatial features from images, capturing important details like edges, textures, and shapes. On the other hand, RNNs excel at handling sequential data, making them well-suited for modeling temporal dependencies or, in this case, dependencies between multiple image labels. By integrating these two powerful architectures, the model leverages the strengths of both CNNs and RNNs to perform better at multi-label classification tasks, where an image may contain multiple relevant objects or concepts. This hybrid framework marks a significant departure from traditional methods that often struggled with multi-label problems due to the complexity of correlating various object labels with the image content[1].

The key contribution of this work lies in the model’s ability to jointly extract visual features and handle multiple labels through a single architecture, making it more efficient than previous approaches that employed separate networks for feature extraction and sequence modeling. Wang et al. demonstrated that their CNN-RNN framework substantially improves accuracy, especially in the context of complex multi-label images, where objects may appear simultaneously in the same scene[1]. This approach not only advances the field of image classification but also sets the stage for future applications in areas like image captioning, where both visual content and the relationships between objects in an image need to be represented effectively.

2.4.2 Deep Learning for Image Description Generation

In their 2020 paper, Sricharitha et al. focused on utilizing deep learning techniques to automatically generate textual descriptions of images. The researchers employed CNNs for extracting features from the input image and Long Short-Term Memory (LSTM) networks for generating coherent textual captions based on those features. This combination allowed the model to capture both the intricate visual details and the linguistic patterns necessary for generating natural, human-like descriptions. The deep learning-based model demonstrated significant improvements over traditional methods that relied on handcrafted features, such as manual object detection or pre-defined descriptors, which were limited in their ability to generate fluent and accurate captions across diverse datasets[2].

The main contribution of this research lies in its demonstration of how deep learning can be effectively applied to image captioning, providing a more accurate and scalable approach to generating descriptive text. The authors found that using CNNs for feature extraction and LSTMs for sequence generation helped the model produce captions that were not only contextually appropriate but also fluent and grammatically correct. By leveraging deep learning, Sricharitha et al. showed that automatic image captioning could move beyond simple object identification and approach a more nuanced understanding of the image's content[2]. This work is particularly important for multilingual image captioning applications, as the deep learning framework can easily be adapted to generate captions in different languages without losing quality or coherence.

2.4.3 Automatic Image Captioning with Soft Attention

The 2017 paper by Chu et al. introduced an innovative approach to image captioning by incorporating soft attention mechanisms in conjunction with ResNet50 and LSTM networks. Attention mechanisms allow the model to selectively focus on specific parts of an image when generating captions, ensuring that the most relevant visual features are emphasized in the description. In this case, the authors used ResNet50, a deep convolutional network known for its exceptional ability to extract high-level features from images, as the feature extractor. LSTM networks, enhanced with soft attention, were then used to generate sequences of text that describe the image. The attention mechanism in this architecture ensures that the model can allocate its "focus" on different parts of the image, depending on the content that is most relevant for the caption at each step of the generation process[3].

Chu et al.'s work made a significant contribution to the field of image captioning by showing that incorporating soft attention mechanisms could dramatically improve the quality and relevance of the captions generated by the model. By allowing the model to focus on the most critical elements of the image, soft attention helps pre-

vent irrelevant or generic descriptions and leads to more accurate and contextually appropriate captions[3]. The ability to visualize attention maps also enhances the interpretability of the model, allowing users to understand which image regions are influencing the captioning process. This approach not only improves caption quality but also contributes to the development of more transparent and interpretable AI systems.

2.4.4 Deep Residual Learning

He et al. (2016) introduced deep residual networks (ResNets), a breakthrough in deep learning that addresses the issue of vanishing gradients, which often hampers the training of very deep neural networks. The key innovation of ResNets lies in their use of "skip connections," which allow information to bypass certain layers in the network, thus preserving gradients and enabling more efficient training of deeper networks. This innovation is particularly significant in the context of image recognition tasks, where deeper networks are required to capture increasingly abstract and high-level features. By enabling networks to be much deeper, ResNets improve feature extraction capabilities, which is crucial for tasks like image captioning, where precise feature representation is needed for generating accurate descriptions[4].

The contribution of He et al.'s work is foundational for understanding how deep residual networks can be applied to various image processing tasks, including image captioning. By introducing ResNets, they significantly enhanced the ability of AI models to extract features from images, which is a critical step in caption generation. The deeper networks allowed for more complex and accurate representations of the image's content, leading to better performance in both image recognition and captioning tasks[4]. ResNets have since become a core component of many advanced image captioning systems, particularly when paired with architectures like LSTMs or attention mechanisms that rely on high-quality visual features to generate detailed and relevant captions.

2.4.5 Explainable AI in Model Interpretability

Aitken et al. (2020) explored the importance of model interpretability in the context of AI systems used for tasks like image captioning, where trust and transparency are crucial. Their research focused on enhancing the interpretability of encoder-decoder architectures through the integration of attention mechanisms. Encoder-decoder models are widely used in tasks such as machine translation and image captioning, as they can efficiently map input data (such as an image) to output sequences (like captions). However, one challenge with these models is that they are often viewed as "black boxes," making it difficult for users to understand how the model arrives at a particular decision[5]. Aitken et al. proposed that incorporating attention mechanisms

could improve the transparency of these models by visualizing the parts of the image the model is focusing on when generating each word of the caption.

The main contribution of this research is the emphasis on explainable AI techniques, which help bridge the gap between model performance and user trust. By incorporating attention mechanisms, Aitken et al. showed that AI systems could not only perform better but also offer insights into their decision-making processes, enhancing their credibility[5]. This is particularly important in applications like medical diagnostics or autonomous vehicles, where understanding how AI models make decisions can be a matter of safety. The work also laid the groundwork for future image captioning systems that aim to be more transparent and accountable, enabling users to better verify and validate the predictions made by AI models.

2.4.6 Using Artificial Tokens for Multilingual Caption Generation

In their 2020 study, Tsutsui and Crandall proposed a novel methodology for multilingual image captioning by using artificial tokens to control language selection during the caption generation process. The idea behind this approach is to integrate artificial tokens into the input sequence, which specify the desired language for the generated caption[6]. This allows the same model to generate captions in different languages, while maintaining the context and relevance of the description. The model is trained with these language-specific tokens, which guide the generation process and help the model switch between languages seamlessly, without compromising the quality or coherence of the generated captions.

The key contribution of this work is the introduction of a method that enables more efficient and accurate multilingual caption generation. By using artificial tokens to control the language output, the model avoids the pitfalls of traditional methods, which often required training separate models for each language or were prone to errors when switching between languages. Tsutsui and Crandall’s approach allows for a more scalable and flexible solution to multilingual image captioning, which is especially valuable in global applications where captions may need to be generated in multiple languages[6]. This work also opens up avenues for further research into language-aware image captioning systems that can seamlessly integrate multilingual capabilities without sacrificing performance or contextual relevance[6].

2.4.7 Transfer Learning and Ensemble Techniques

Jiazhi Liang (2020) focused on the application of transfer learning and ensemble techniques to improve the performance of image captioning systems. Transfer learning involves using a pre-trained model on a large dataset and fine-tuning it for a specific task, which can significantly reduce the amount of data and computational resources

required for training. In their study, Li et al. demonstrated that by combining pre-trained networks, such as ResNet50, with ensemble learning approaches, they were able to improve the detection of various conditions within images[7]. Ensemble methods combine the predictions of multiple models to enhance overall performance and reduce the risk of errors that could occur with individual models. This hybrid approach allowed for more robust image captioning systems capable of handling a wide range of images and conditions[7].

The main contribution of this paper is the combination of transfer learning with ensemble techniques, which not only reduced the training time but also led to better performance on complex multi-class image classification and captioning tasks[7]. By leveraging the knowledge embedded in pre-trained models and combining it with multiple diverse models, Li et al. showed that the performance of image captioning systems could be significantly improved, particularly in cases where training data is limited or where the model needs to generalize to a variety of different image types. This approach highlights the potential of hybrid models in creating more accurate, efficient, and versatile systems for generating captions and other forms of image analysis[7].

Title	Authors	Key Contributions
CNN-RNN: A Unified Framework for Multi-label Image Classification	Jiang Wang, Yi Yang, Junhua Mao, Zhiheng Huang, Chang Huang, Wei Xu (2016)	Combined CNNs and RNNs for robust spatial-temporal modeling.
Image Description Generator Using Deep Learning	Sricharitha, S.V. Amit, Md B Sultan Bahyal (2023)	Integrated CNNs and LSTMs for descriptive captioning.
Automatic Image Captioning Based on ResNet50 and LSTM with Soft Attention	Yan Chu et al. (2020)	ResNet50 and LSTM with soft attention for focused captions.
Deep Residual Learning for Image Recognition	Kaiming He, Xiangyu Zhang, Shaoqing Ren, Jian Sun (2016)	Introduced ResNet for enhanced feature extraction.
Understanding How Encoder-Decoder Architectures Attend	Kyle Aitken, Vinay V. Ramasesh, Yuan Cao, Niru Maheswaranathan (2021)	Improved interpretability with attention mechanisms.
Using Artificial Tokens to Control Languages for Multilingual Image Caption Generation	Satoshi Tsutsui and David Crandall (2017)	Managed multilingual captioning via artificial tokens.
Image classification based on RESNET	Jiazhi Liang (2020)	Enhanced performance with pre-trained and ensemble models.

Table 2.2: Literature Review Summary

Chapter 3

METHODOLOGY

3.1 Overview

The Multilingual Image Caption Generator is an advanced AI system designed to generate meaningful textual descriptions for images in multiple languages. The project integrates cutting-edge computer vision and natural language processing techniques, leveraging a combination of pre-trained models and customized deep learning architectures. ResNet50, a convolutional neural network pre-trained on ImageNet, serves as the encoder to extract high-dimensional features from input images. These features are then passed to an LSTM-based decoder, which sequentially generates captions. The inclusion of artificial tokens enables the model to handle multilingual captioning by specifying the target language during both training and inference phases. This approach ensures a unified architecture capable of producing linguistically and contextually accurate captions without requiring separate models for each language.

To enhance accessibility, the system incorporates audio descriptions using the Google Text-to-Speech (gTTS) API, converting textual captions into speech. The project follows a systematic methodology: starting with the preparation and preprocessing of the Flickr30k dataset, which includes diverse real-world image scenarios with detailed annotations. Training involves the use of categorical cross-entropy loss and the Adam optimizer, with the model evaluated on metrics such as BLEU scores to ensure caption quality. Finally, a Flask-based web interface allows users to upload images, generate captions, select their desired language, and access audio descriptions. This phased and modular approach ensures that the system is robust, scalable, and adaptable to diverse real-world applications, making it a powerful tool for accessibility, multimedia management, and human-computer interaction.

3.2 Data Collection

Data collection is the process of gathering and preparing relevant datasets to train, validate, and test machine learning models. Proper data collection ensures the model has sufficient examples to learn patterns, handle diverse scenarios, and generalize

effectively. The quality and diversity of the data directly impact the model’s ability to generate accurate, contextually relevant, and linguistically diverse captions. A well-curated dataset lays the foundation for building a robust and reliable multilingual image captioning system.

3.3 Dataset: Flickr30k

The Flickr30k dataset is a highly regarded benchmark in the field of image captioning, renowned for its richness and diversity in both visual content and annotations. Comprising 30,000 high-quality images, the dataset captures a wide array of real-world scenarios, including urban settings, natural landscapes, human activities, and interactions with animals and objects. Each image is annotated with five human-generated captions, providing a comprehensive linguistic perspective on the visual content. These captions are designed to emphasize different aspects of the image, such as actions, objects, relationships, and contextual elements, making Flickr30k an invaluable resource for training and evaluating models that aim to generate descriptive and contextually relevant captions.



Figure 3.1: Sample Flickr30k images

3.3.1 Reasons for Selecting Flickr30k

1. Rich Captions

Each image in Flickr30k is annotated with five human-generated captions, offering diverse linguistic perspectives for the same visual content. This variety helps models understand different ways to describe similar scenes, such as focusing on objects, actions, or overall context. For example, an image of a dog playing fetch may have captions emphasizing "a dog," "a ball," or "a game." This diversity is crucial for training robust models that can generalize well across unseen data, making the dataset ideal for image-to-text learning tasks.

2. Wide Range of Scenarios

Flickr30k includes images of people, animals, objects, natural landscapes, urban settings, and activities, providing a comprehensive representation of real-world scenarios.

This variability ensures that the trained models can handle diverse image content and complex interactions. By exposing the model to a wide array of scenes during training, it becomes better equipped to understand and describe the nuances of complex visual environments.

3. Standard Benchmark for Comparisons

Flickr30k is widely recognized as a standard benchmark in academic and industrial research on image captioning. Its extensive use in the community allows researchers to compare the performance of their models with existing state-of-the-art methods. This benchmark status provides a common ground for evaluating and validating the effectiveness of new architectures, such as your ResNet50-LSTM-based multilingual caption generator.

4. Suitability for Multilingual Tasks

Although the dataset's original annotations are in English, its rich content makes it ideal for extending into multilingual settings. By leveraging artificial tokens to denote language or integrating translation APIs, captions can be easily adapted for multiple languages. This flexibility allows the Flickr30k dataset to serve as the foundation for training models capable of generating captions in multiple languages, addressing diverse linguistic and cultural needs.

5. Human-Centric Annotations

The captions in Flickr30k are written by human annotators, ensuring that they are descriptive, contextually accurate, and aligned with real-world interpretations. Unlike machine-generated captions, human annotations include a natural linguistic flow and diverse sentence structures. This provides the model with high-quality training data that aligns closely with user expectations.

6. Practical Applications

The dataset's variety and richness make it applicable to real-world scenarios, such as accessibility tools, multimedia content management, and e-commerce platforms. By training on Flickr30k, models gain the capability to generate captions that are relevant to a wide range of applications, enhancing their usability and impact.

7. Scalability and Integration with Other Datasets

Flickr30k can be seamlessly integrated with other datasets, such as MS COCO or Pascal VOC, for larger-scale training. This compatibility allows researchers to combine datasets to further improve the robustness and generalization of their models.

3.3.2 Dataset Structure

The Flickr30k dataset provides a comprehensive pairing of images and human-generated captions, making it a rich resource for training image captioning systems. Each of

the 30,000 images is accompanied by five unique captions, which describe the image content from different perspectives. This diversity in annotations allows models to learn various ways of interpreting the same scene, thereby improving their ability to generalize across unseen data.

Image-Caption Pairing

Each image is annotated with captions that focus on distinct aspects of the scene. The captions typically focus on four key aspects:

- **Objects:** Captions that identify and describe specific items within the image, such as people, animals, or objects. For example, a caption like “a red car parked on the street” focuses on the object (the car) and its attributes (color and location). This helps the model learn to recognize and describe objects in diverse settings.
- **Actions:** These captions describe activities or movements occurring in the scene. For instance, “a boy kicking a soccer ball” provides details about an ongoing action, allowing the model to learn how to capture dynamic events and actions that are integral to understanding an image.
- **Settings:** Some captions emphasize the environment or background of the image. For example, “a park with green trees” provides context about the location and atmosphere, helping the model understand the broader setting in which objects or people are situated. Recognizing and describing the environment is essential for producing contextually appropriate captions.
- **Relationships:** These captions describe the interactions or relationships between objects, people, or both. For example, “a mother holding her baby” goes beyond describing individual entities and focuses on the relationship between them. This teaches the model to understand complex interactions and spatial relationships within an image.

Caption Diversity

One of the most valuable features of the Flickr30k dataset is the diversity of the captions for each image. By providing five unique captions for every image, the dataset ensures that the model has access to a range of linguistic styles, phrasing, and descriptive approaches. This diversity is not only important for improving the variety of the generated captions but also for generalization—helping the model to understand different ways of describing the same image.

Image ID	Caption 1	Caption 2	Caption 3	Caption 4	Caption 5
1000092795.jpg	Two young guys with shaggy hair look at their hands while hanging out in the yard.	Two young, White males are outside near many bushes.	Two men in green shirts are standing in a yard.	A man in a blue shirt standing in a garden.	Two friends enjoy time spent together.
10002456.jpg	Several men in hard hats are operating a giant pulley system.	Workers look down from up above on a piece of equipment.	Two men working on a machine wearing hard hats.	Four men on top of a tall structure.	Three men on a large rig.
1000268201.jpg	A child in a pink dress is climbing up a set of stairs in an entry way.	A little girl in a pink dress going into a wooden cabin.	A little girl climbing the stairs to her playhouse.	A little girl climbing into a wooden playhouse.	A girl going into a wooden building.

Table 3.1: Sample Image-Caption Pairing

- **Variations in sentence structure:** Different sentence structures are used to describe the same scene, allowing the model to learn to produce more flexible outputs. For instance, “a dog playing in the yard” might be rephrased as “in the yard, a dog is playing,” providing two different ways of constructing a description while focusing on the same content. This trains the model to generate captions that reflect various syntactic choices.
- **Use of synonyms or alternate expressions:** The dataset also includes variations in vocabulary, using synonyms or alternate expressions for the same concept. For example, one caption might say “riding a bike,” while another might use “cycling.” This variation helps the model to better understand language nuances and generate captions that are not only syntactically accurate but also semantically rich.
- **Context-specific details:** Each caption might highlight different elements of the image based on what the annotator chooses to focus on. For example, a caption may mention the people in the image, while another might emphasize the background scenery or the objects present. This variability encourages the model to generate captions that can capture a broad range of details depending on the context.

Image Variety

- **People-Centric:** These images feature individuals or groups engaged in various activities, ranging from sports and recreation to everyday actions like walking, talking, or interacting with each other. The variety of activities captured, including both dynamic actions (e.g., running, jumping) and more static interactions (e.g., sitting, talking), allows models to learn how to describe a wide array of human behaviors. These images help train systems to understand human-centered contexts, actions, and interactions, which are often key elements in generating accurate captions.
- **Animal-Centric:** The dataset includes a significant number of images featuring both domestic pets and wild animals. These images are captured in diverse settings, from natural environments (e.g., forests, savannahs, or jungles) to urban or human-made environments (e.g., parks or zoos). The wide range of animals featured, along with their various activities—whether they’re interacting with people or other animals or simply existing in their habitat—helps models learn to recognize animals, understand their behaviors, and generate appropriate captions. Additionally, it allows models to make contextual decisions based on the environment in which the animals are depicted.
- **Object-Centric:** Many images in the dataset focus on objects—such as cars, bicycles, or household items—often in combination with human or animal activity. These images help train models to identify and describe inanimate objects in context, either alone or when they interact with other objects or people. Descriptions of such objects might also involve their functional relationships, like a person riding a bike or a child playing with a toy. By learning to recognize these everyday items and their roles in various scenarios, image captioning systems become better at producing descriptive text for images focused on objects.
- **Settings and Landscapes:** This category includes images depicting various environments and landscapes, ranging from indoor settings (e.g., living rooms, kitchens, or offices) to outdoor landscapes (e.g., parks, beaches, forests, or urban streets). The images in this category emphasize the background or setting of the image, helping models to understand the context of the scene. For example, a caption might describe the scenic elements of a park, the layout of an urban street, or the ambience of a beach. These settings add valuable context to the understanding of objects and actions within the scene, allowing for more holistic caption generation.

3.4 Data Preprocessing

Data preprocessing is a critical step in any machine learning task, involving the preparation of both images and captions to ensure compatibility with machine learning algorithms. This section outlines the methodology employed to process the Flickr30k dataset, which consists of 30,000 images, each accompanied by five human-generated captions. The goal of the data processing pipeline is to clean and standardize the textual captions while resizing and normalizing the images, enabling seamless integration into the deep learning model.

3.4.1 Image Preprocessing

The images in the Flickr30k dataset exhibit diverse sizes, resolutions, and formats. To ensure consistency and compatibility with the ResNet50 model used as the encoder, a preprocessing pipeline was applied to standardize the images.

The preprocessing steps are as follows:

1. Resizing

All images were resized to a fixed dimension of 224x224 pixels, the input size required by the ResNet50 model. Resizing ensures uniformity across the dataset and eliminates inconsistencies that may arise from varying image dimensions. This step is crucial for enabling batch processing and maintaining model efficiency.

2. Normalization

Pixel values of the resized images were normalized to the range $[0, 1]$. This was achieved by dividing each pixel value by 255, the maximum possible value for an 8-bit pixel. Normalization is a common practice in deep learning as it helps in faster convergence during model training by ensuring that the input data is scaled to a consistent range. It also reduces the potential for numerical instability during computation.

3. Conversion to Arrays

The processed images were converted into numerical arrays using the OpenCV library. This step transforms the visual content of each image into a format suitable for input into machine learning models. The arrays are structured in a 4D tensor format (batch size, height, width, channels), which is compatible with the ResNet50 model's input requirements.

4. Augmentation

To increase the diversity of the dataset and improve the robustness of the model, optional image augmentation techniques such as flipping, rotation, and brightness adjustment can be applied. This ensures that the model generalizes better to unseen data by simulating real-world variations in image appearance.

3.4.2 Caption Preprocessing

The textual captions corresponding to the images needed extensive preprocessing to make them compatible with the deep learning pipeline. Captions are textual data, which must be converted into numerical representations for training.

Steps for Caption Preprocessing:

1. Lowercasing

All captions were converted to lowercase to ensure uniformity and eliminate case sensitivity. This step ensures that variations in letter casing (e.g., "Boy" vs. "boy") do not result in separate entries in the vocabulary.

2. Removing Special Characters and Numbers

Punctuation, special characters, and numbers were stripped from the captions to minimize noise. This cleaning process ensures that the dataset focuses on meaningful words, such as nouns, verbs, and adjectives, which are essential for generating descriptive captions. Words like "soccer", "dog", and "tree" retain their semantic importance, while irrelevant symbols are removed.

3. Adding Artificial Tokens

To help the model understand the structure of each caption, two special tokens, `startseq` and `endseq`, were added. These tokens denote the beginning and end of each caption, guiding the sequence-to-sequence architecture during training and inference.

- **Example**

Original Caption: "A boy playing soccer in the park."

Processed Caption: "`startseq` boy playing soccer park `endseq`"

4. Tokenization

Tokenization converts words into numerical representations. Each word in the captions was assigned a unique integer ID based on its frequency in the dataset. Frequently occurring words were assigned lower IDs, while less common words received higher IDs. This mapping simplifies the captions into sequences of integers, making them compatible with machine learning models.

- **Example**

Processed Caption: "`startseq` boy playing soccer park `endseq`"

Tokenized Caption: [1, 2, 3, 4, 5, 6]

Here, the words "boy", "playing", "soccer", and "park" are assigned unique integer IDs, along with `startseq` and `endseq`

5. Vocabulary Creation

The vocabulary was restricted to the top 5,000 most frequently occurring words in the dataset to balance computational efficiency with linguistic richness. Words outside

this vocabulary were replaced with a special token UNK (unknown), ensuring that rare or irrelevant terms do not dominate the learning process.

3.4.3 Data Cleaning

The data cleaning process is a crucial step to ensure that the captions are consistent, noise-free, and suitable for training the deep learning model. By applying various cleaning techniques, the captions are standardized, enhancing the model's ability to learn effectively from the data.

1. Lowercasing

All captions were converted to lowercase to ensure uniformity and eliminate case sensitivity. This prevents the model from treating the same word with different capitalizations (e.g., "Dog" and "dog") as separate entities.

2. Removing Non-Alphabetic Characters

Punctuation marks, numbers, and other non-alphabetic characters were removed to reduce noise in the data. This step ensures that the captions consist of meaningful words, such as nouns, verbs, and adjectives, which contribute significantly to image description.

3. Whitespace Normalization

Multiple spaces, tabs, or other redundant whitespace were replaced with a single space. This step avoids formatting inconsistencies and ensures that the text is clean and well-structured.

4. Stopword Retention

Commonly used words like "a," "the," and "is" were retained in the captions. While these words are often removed in other NLP tasks, they provide grammatical structure and coherence, which is essential for generating descriptive captions.

5. Handling Special Cases

Words with apostrophes, such as "it's" or "don't," were carefully handled to ensure they are appropriately tokenized without altering their meaning. This prevents loss of semantic information.

3.4.4 Exploratory Data Analysis

An in-depth analysis of the captions' vocabulary was performed to understand the textual content of the dataset better. This exploratory data analysis (EDA) focused on examining word frequency distributions and visualizing the insights through bar charts, enabling us to identify common patterns and rare occurrences in the dataset.

Word Frequency Analysis

The captions were tokenized by splitting them into individual words. Each word's frequency was calculated to determine how often it appeared across all captions in the dataset. This analysis provided valuable insights into the vocabulary, revealing commonly described objects, actions, and attributes in the images.

The word frequency distribution not only highlights the dataset's linguistic richness but also helps identify redundant or irrelevant terms that can be excluded to enhance the model's training efficiency.

Visualization of Word Frequencies To better illustrate the word frequency analysis, bar charts were created to visualize the most and least frequently appearing words in the captions

- **Top 50 Most Frequent Words:**

The most common words in the captions are often related to objects, actions, or attributes frequently captured in the images. These words form the core vocabulary that the model will heavily rely on during training. The bar chart below displays the distribution of the top 50 frequently appearing words, emphasizing their prevalence in the dataset.

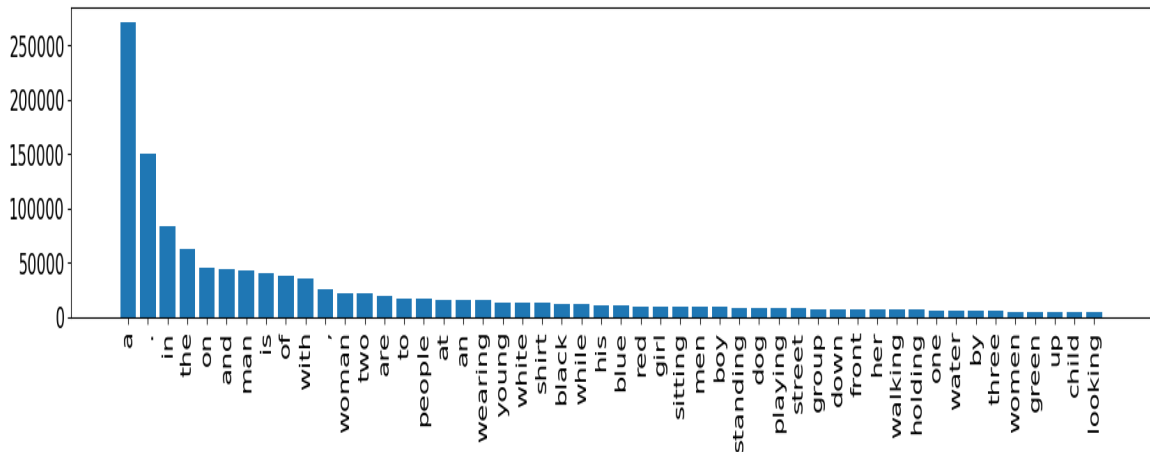


Figure 3.2: Top 50 Most frequently appearing words

- **Least 50 Frequent Words:**

The least frequent words often include rare or specialized terms that might represent unique scenarios or objects captured in a small subset of the images. While these words are less critical to the model's performance, they contribute to the dataset's diversity. The bar chart below shows the distribution of the least frequently appearing words, providing insights into the dataset's linguistic variability.

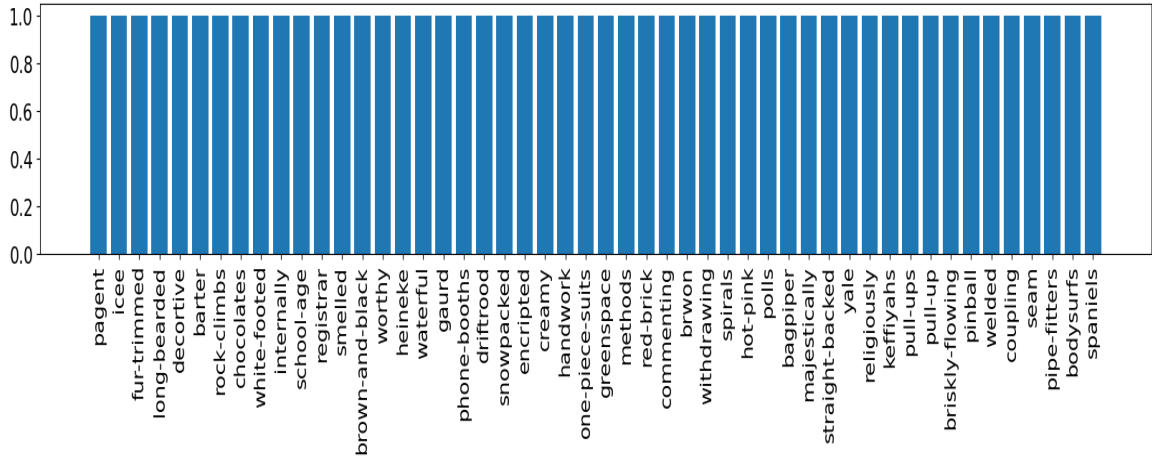


Figure 3.3: Least 50 appearing words

3.4.5 Organizing Captions by Image

The captions are grouped and organized by their corresponding image IDs to enable efficient retrieval and management of multiple descriptions for a single image. Each image can have multiple captions, capturing different aspects or details of the same scene. This is crucial in datasets used for training image captioning models, where diverse descriptions provide varied learning examples.

The dataset is structured as a dictionary, where:

- **Key:** The unique Image ID (without the file extension). This serves as an identifier for each image and allows the grouping of its captions.
- **Value:** A list of captions associated with the image. Each entry in this list contains a textual description that describes the content or details of the image.

3.4.6 Storing Cleaned Data

Once the captions were cleaned, tokenized, and organized by image ID, the next step was to store this processed data in a format that would be easily accessible and ready for use in machine learning models. The cleaned data was saved in a new file, tokens-clean.txt, which contains the final set of captions grouped by their respective image IDs.

3.5 Model Architecture

The model architecture for the Multilingual Image Caption Generator consists of two main components: the encoder and the decoder. The encoder extracts visual features from images, while the decoder generates textual descriptions in a sequential manner.

This combination of a convolutional neural network (CNN) as the encoder and a recurrent neural network (RNN) as the decoder forms the backbone of the system.

3.5.1 Convolutional Neural Network (CNN)

Convolutional Neural Networks (CNNs) are a class of deep neural networks that have proven highly effective in computer vision tasks, such as image classification, object detection, and image captioning. CNNs are designed to automatically and adaptively learn spatial hierarchies of features from input images. They operate by applying convolutional layers that scan the image through small filters (or kernels), detecting patterns like edges, textures, and more complex structures as they move through deeper layers.

Components of CNN:

- 1. Convolutional Layers:** These layers apply a series of filters to the input image. Each filter extracts specific features such as edges, corners, or textures. As the image is processed through multiple layers, the filters begin to detect more complex and abstract features, ultimately capturing the high-level patterns necessary for accurate image interpretation.
- 2. Activation Function (ReLU):** After the convolutional operation, the output is passed through an activation function, often ReLU (Rectified Linear Unit), which introduces non-linearity. This helps the network model more complex relationships in the data.
- 3. Pooling Layers:** To reduce the spatial dimensions of the image while retaining important features, CNNs often employ pooling layers, such as max pooling. These layers down-sample the feature maps, reducing computational complexity and helping the model generalize better by making it less sensitive to small translations in the image.
- 4. Fully Connected Layers (Dense Layers):** In the final layers of the CNN, the extracted features are flattened into a 1D vector and passed through fully connected layers. These layers make the final predictions or pass the features to the next part of the model, in this case, the RNN or decoder, for further processing.
- 5. Output Layer:** The output layer produces a feature vector that represents the entire image, which can then be used for tasks like classification or, in the case of image captioning, as input for the next stage of the model (the RNN).

Role of CNN in Image Captioning:

In image captioning tasks, CNNs are responsible for transforming the input image into a compact representation, a feature vector, that encodes the most important visual

information. These feature vectors are then passed to the RNN (or decoder), which generates the caption based on the extracted features.

1. Feature Extraction: The CNN extracts relevant visual features, such as objects, textures, and spatial relationships between elements within the image. These features are crucial for generating accurate captions, as they allow the captioning model to understand the content of the image.

2. Pre-trained Networks: Often, CNNs like ResNet50 or VGGNet are pre-trained on large datasets (like ImageNet) and then fine-tuned for the specific image captioning task. This transfer learning helps the model learn from a vast amount of data before being applied to a specific task, thus improving performance and reducing training time.

3.5.2 Recurrent Neural Network (RNN)

A Recurrent Neural Network (RNN) is a class of artificial neural networks designed to process sequential data. Unlike traditional feedforward neural networks, RNNs have connections that form cycles within the network, which allows them to maintain an internal state or memory. This ability to "remember" previous inputs makes RNNs particularly well-suited for tasks involving sequences, such as speech recognition, language modeling, and image captioning.

Key Characteristics of RNNs:

- **Sequential Processing:** RNNs are designed to handle sequential data, where the order of the input matters. Each element in the sequence is processed one at a time, and the network maintains an internal state that evolves as new data points are processed.
- **Hidden State (Memory):** At each time step, the RNN processes an input (e.g., a word or image feature vector) and updates its hidden state. This hidden state is used to store information about previous inputs in the sequence, which helps the RNN maintain context and capture temporal dependencies. This characteristic makes RNNs especially powerful for tasks like language modeling, where context from earlier words is crucial for generating accurate predictions.
- **Weight Sharing:** RNNs share weights across time steps, meaning that the same parameters are used to process different elements of the sequence. This weight-sharing property allows RNNs to generalize well to sequences of varying lengths and learn patterns across time.

Role of RNN in Image Captioning:

In the context of image captioning, the role of the RNN is to generate textual descriptions based on the features extracted from the image. After the image features

have been processed by a Convolutional Neural Network (CNN), the resulting feature vector is passed to the RNN, which decodes the image representation into a coherent sequence of words that form a caption.

- **Text Generation:** The RNN generates captions by predicting one word at a time, using the context provided by the feature vector from the CNN and its internal hidden state. As each word is generated, the RNN updates its internal state to capture the evolving context of the sentence.
- **Sequential Nature of Language:** Since language is inherently sequential (i.e., the meaning of a sentence depends on the order of the words), RNNs are ideal for generating captions word by word, ensuring that each word is influenced by the ones that came before it. The output of the RNN at each time step is typically passed through a softmax layer to select the most probable word from the vocabulary.
- **Handling Complex Sentences:** More advanced versions of RNNs, such as LSTM and GRU, help improve the ability of the model to learn longer-term dependencies within the sequence, allowing the image captioning system to generate more complex and grammatically correct sentences.
- **Multilingual Captioning:** In multilingual image captioning tasks, RNNs can be extended to generate captions in different languages. By training the RNN on datasets containing captions in multiple languages, the model can generate captions in the desired language, leveraging the same underlying structure while accounting for language-specific nuances.

3.5.3 Encoder

An encoder is a fundamental component in many machine learning and deep learning architectures. Its primary purpose is to transform input data into a compressed, meaningful, and often lower-dimensional representation, which can then be processed further by other parts of the system. Encoders are commonly used in tasks like sequence modeling, image processing, and natural language processing.

Purpose of an Encoder

- **Computer Vision:** Encoders extract features from images in tasks like object detection or image segmentation.
- **Sequence-to-sequence tasks (e.g., machine translation):** Encoders process source language sentences and produce a representation that decoders use to generate translations.
- **Autoencoders:** Encoders compress input data to a latent space for dimensionality reduction or feature extraction.

- Transformers: Encoders process input sequences to produce contextualized embeddings in models like BERT and GPT.

3.5.3.1 Resnet50

ResNet-50 is one of the most well-known deep convolutional neural networks (CNNs) and belongs to the ResNet (Residual Networks) family of models. It was introduced in the groundbreaking paper "Deep Residual Learning for Image Recognition" by He et al. in 2015. ResNet-50 is a 50-layer network that leverages residual learning to overcome challenges that arise from training very deep networks.

ResNet-50, despite its relatively "moderate" depth (compared to even deeper ResNet architectures like ResNet-101 or ResNet-152), achieves exceptional performance on a variety of vision tasks. Its architecture allows it to train deeper networks without facing the problems of vanishing gradients or degradation (which usually occurs in deeper networks).

Key Concepts in ResNet-50

1. Residual Connections:

- The hallmark of the ResNet architecture is the introduction of residual connections, which help mitigate the issue of vanishing gradients. A residual connection is simply a shortcut that skips one or more layers, allowing the input to bypass them and be added to the output of those layers. This allows the network to learn the difference (residual) between the desired output and the input, rather than learning the output directly.
- Residual Learning: Instead of learning the output of a layer directly, ResNet learns the residual (i.e., the difference). This has been shown to ease training in deep networks and enable the construction of much deeper models.

2. Bottleneck Architecture:

- ResNet-50 uses a bottleneck architecture, where each block consists of three convolutional layers instead of two. This reduces the number of computations compared to the original ResNet, while maintaining performance. The three layers in each block are:
 - 1x1 Convolution: Reduces the dimensionality (i.e., compresses the feature map).
 - 3x3 Convolution: Performs the main feature extraction.
 - 1x1 Convolution: Restores the dimensionality.
- This structure reduces the overall number of parameters and allows the network to be trained more efficiently.

3. Deep Learning with Residuals:

- Deep neural networks traditionally suffer from vanishing gradients, where gradients become very small as they propagate backward through many layers. This makes it hard to update weights in earlier layers. ResNet mitigates this problem by allowing gradients to flow directly through the residual connections, making it possible to train very deep networks (hundreds of layers) efficiently.

ResNet-50 Architecture

ResNet-50 is a deep convolutional neural network (CNN) that addresses the vanishing gradient problem through residual connections (skip connections). These connections allow gradients to propagate effectively through deeper networks, enabling the model to learn more complex features without degrading performance. The ResNet-50 model contains 50 layers, which makes it an excellent choice for tasks like image classification, object detection, and segmentation.

The architecture of ResNet-50 is composed of several key components

- **Input Layer:**

The model begins with an input layer that receives an image of size 224x224x3 (height, width, and 3 color channels - RGB). This input is the starting point of the network, where the image data is fed into the model.

- **Initial Convolution and Pooling:**

After the input layer, a ZeroPadding2D layer is applied, adding padding around the image to maintain the spatial dimensions after the convolution operations. This step increases the size of the image slightly.

The image is then passed through a 7x7 Convolution layer with 64 filters and a stride of 2, which extracts initial features like edges and textures. The output of this layer is a feature map with reduced spatial dimensions.

A MaxPooling2D layer with a 3x3 filter and stride 2 follows to further reduce the spatial dimensions of the feature map, making the network more efficient and reducing computational complexity.

- **Residual Blocks:**

The key feature of the residual block is the skip connection, where the input to the block is directly added to the output of the block using an Add layer. This helps in the training of deep networks by mitigating the vanishing gradient problem.

- **Batch Normalization and ReLU Activation:**

After each convolutional operation in the residual block, BatchNormalization is applied to standardize the activations, ensuring that they have a mean of 0

and a standard deviation of 1. This stabilizes the training and helps the model converge faster.

The ReLU Activation function is applied to introduce non-linearity after each convolution. This is crucial because it allows the network to learn complex, non-linear relationships in the data.

- **Global Average Pooling:**

After passing through all the residual blocks, the network applies Global Average Pooling. This operation reduces each feature map to a single value by averaging all the values in the feature map. This significantly reduces the spatial dimensions while retaining the most important features, creating a compact representation of the learned features.

- **Fully Connected Layer (Dense Layer):**

The output of the Global Average Pooling is flattened into a single vector and passed through a fully connected (dense) layer. The dense layer is used to make the final predictions, such as class probabilities for classification tasks. For example, in the case of ImageNet classification, the final layer may contain 1000 units (one for each class).

Model: "resnet50"

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 224, 224, 3)	0	-
conv1_pad (ZeroPadding2D)	(None, 230, 230, 3)	0	input_layer[0][0]
conv1_conv (Conv2D)	(None, 112, 112, 64)	9,472	conv1_pad[0][0]
conv1_bn (BatchNormalization)	(None, 112, 112, 64)	256	conv1_conv[0][0]
conv1_relu (Activation)	(None, 112, 112, 64)	0	conv1_bn[0][0]
pool1_pad (ZeroPadding2D)	(None, 114, 114, 64)	0	conv1_relu[0][0]
pool1_pool (MaxPooling2D)	(None, 56, 56, 64)	0	pool1_pad[0][0]
conv2_block1_1_conv (Conv2D)	(None, 56, 56, 64)	4,160	pool1_pool[0][0]
.....			
conv5_block3_add (Add)	(None, 7, 7, 2048)	0	conv5_block2_out... conv5_block3_b...
conv5_block3_out (Activation)	(None, 7, 7, 2048)	0	conv5_block3_add...
avg_pool (GlobalAveragePool...)	(None, 2048)	0	conv5_block3_out...
predictions (Dense)	(None, 1000)	2,049,000	avg_pool[0][0]

Total params: 25,636,712 (97.80 MB)

Trainable params: 25,583,592 (97.59 MB)

Non-trainable params: 53,120 (267.50 KB)

Figure 3.4: Summary of ResNet-50 Architecture

How ResNet-50 Works in Practice

In practice, the ResNet-50 architecture can be divided into three main stages:

1.Feature Extraction:

The image is passed through the initial convolutional layers and multiple residual blocks to extract high-level features from the image. Each block learns complex patterns, such as shapes, textures, and objects.

2.Global Average Pooling:

The output from the last residual block undergoes global average pooling to reduce the dimensionality of the feature map to a fixed size. This step captures the most relevant features from the entire image.

3.Classification:

The final feature vector is passed through a fully connected layer (usually followed by a softmax function) to produce the final class probabilities, making ResNet-50 highly suitable for image classification tasks.

3.5.4 Decoder

In deep learning, the decoder is a component commonly used in sequence-to-sequence models, such as in tasks like machine translation, image captioning, and image segmentation. The decoder's primary function is to generate an output sequence (or image, in the case of segmentation tasks) from a learned representation of the input data. This is typically the second part of an encoder-decoder architecture, where the encoder processes the input and compresses it into a fixed-size representation, which is then expanded or transformed by the decoder into the final output.

The decoder is built upon an LSTM (Long Short-Term Memory) network, a specialized type of recurrent neural network (RNN) designed to handle sequential data effectively.

3.5.4.1 Long Short-Term Memory (LSTM)

Long Short-Term Memory (LSTM) is a type of Recurrent Neural Network (RNN) that was specifically designed to address the challenges of learning long-term dependencies in sequential data. Traditional RNNs, while effective for sequential tasks, suffer from a limitation known as the vanishing gradient problem, where gradients (used for updating weights during training) become very small, making it difficult for the network to learn from distant time steps. LSTMs were introduced by Sepp Hochreiter and Jürgen Schmidhuber in 1997 to overcome this issue and improve the ability of RNNs to learn and remember long-term dependencies.

Key Features of LSTM:

1. Memory Cells:

The core of an LSTM unit is the memory cell, which is responsible for storing information over time. This memory cell allows the LSTM to "remember" important information for longer periods, unlike traditional RNNs, where information tends to get lost or overwritten during training.

2. Gates:

LSTMs use gates to control the flow of information into and out of the memory cell. There are three main types of gates in an LSTM:

- **Forget Gate:** Decides what information from the memory cell should be discarded or "forgotten." It looks at the current input and the previous hidden state and outputs a value between 0 and 1, indicating the degree to which the memory should be retained or erased.
- **Input Gate:** Controls what new information should be stored in the memory cell. It looks at the current input and the previous hidden state, and outputs a value between 0 and 1, determining how much of the incoming information will be added to the memory.
- **Output Gate:** Decides what information from the memory cell will be used in the next hidden state. It filters the information from the memory cell and passes it to the next time step or output layer.

3. Cell State:

The cell state acts as the long-term memory of the LSTM. It runs through the entire network and is updated at each time step based on the forget and input gates. The cell state is passed along the sequence, helping the LSTM to preserve important information over long sequences.

4. Hidden State:

The hidden state is the short-term memory of the LSTM and is used as the output at each time step. It is influenced by both the memory cell and the current input.

Role of LSTM in Image Captioning:

In image captioning, the LSTM plays a critical role in generating coherent and contextually relevant captions based on the features extracted from the image by a Convolutional Neural Network (CNN).

1. Context Preservation:

After the CNN extracts high-level features from the image, the LSTM takes these features as input and generates the caption word by word. The LSTM's ability to maintain context over time makes it well-suited for tasks like caption generation,

where the meaning of a sentence depends on the correct sequence of words.

2. Handling Long Dependencies:

LSTMs are capable of capturing long-range dependencies in a sequence of words, which is crucial for generating more complex and grammatically accurate captions. For instance, if the caption needs to describe a relationship between multiple objects in an image (e.g., "A woman is holding a baby while standing next to a dog"), the LSTM can retain the context of earlier words like "woman" and "baby" as it generates later words like "dog" and "standing."

3. Attention Mechanism:

LSTMs can be enhanced with an attention mechanism, which allows the network to focus on specific parts of the image while generating each word in the caption. For example, when generating the word "dog," the attention mechanism directs the LSTM to focus on the part of the image containing the dog. This leads to more accurate and detailed captions, as the model can selectively attend to relevant parts of the image.

4. Multilingual Captioning:

In multilingual image captioning tasks, LSTMs can be trained on datasets with captions in multiple languages. This allows the model to generate captions in different languages while maintaining the same underlying architecture, making LSTMs a versatile choice for cross-lingual tasks.

LSTM Layers in Image Captioning

In image captioning models, Long Short-Term Memory (LSTM) networks typically consist of several key layers: Embedding Layer, LSTM Layer, and Dense Layer. These layers work together to process input sequences and generate meaningful captions.

1. Embedding Layer:

The embedding layer converts words or tokens into dense, continuous vector representations (embeddings). Each word in the vocabulary is mapped to a vector of fixed size, which captures semantic relationships between words. This layer helps the LSTM model work with a more compact representation of the input sequence rather than raw word indices.

2. LSTM Layer:

The LSTM layer processes the embedded words sequentially, capturing temporal dependencies and context over the sequence. It maintains an internal memory (cell state) to remember important information across time steps. As the LSTM processes each word in the sequence, it updates its hidden state, which is used to predict the next word in the caption. Multiple LSTM layers can be stacked for deeper modeling of complex sequences.

3. Dense Layer:

The dense layer is typically used at the end of the LSTM network to map the final hidden state to a probability distribution over the vocabulary. This is done through a softmax function, which generates a probability for each word in the vocabulary. The word with the highest probability is selected as the next word in the caption sequence.

Model: "functional_1"

Layer (type)	Output Shape	Param #	Connected to
input_layer_2 (InputLayer)	(None, 80)	0	-
input_layer_1 (InputLayer)	(None, 2048)	0	-
embedding (Embedding)	(None, 80, 50)	349,450	input_layer_2[0]-
dropout (Dropout)	(None, 2048)	0	input_layer_1[0]-
dropout_1 (Dropout)	(None, 80, 50)	0	embedding[0][0]
dense (Dense)	(None, 512)	1,049,088	dropout[0][0]
lstm (LSTM)	(None, 512)	1,153,024	dropout_1[0][0]
add (Add)	(None, 512)	0	dense[0][0], lstm[0][0]
dense_1 (Dense)	(None, 256)	131,328	add[0][0]
dense_2 (Dense)	(None, 6989)	1,796,173	dense_1[0][0]

Total params: 4,479,063 (17.09 MB)

Trainable params: 4,479,063 (17.09 MB)

Non-trainable params: 0 (0.00 B)

Figure 3.5: LSTM Architecture

3.5.5 Encoder-Decoder framework

The Encoder-Decoder framework is a widely used architecture in deep learning, especially for tasks involving input and output sequences of varying lengths or types. The encoder processes the input data and compresses it into a fixed-size context vector or latent representation, capturing the essential features. The decoder then takes this representation and generates the output sequence. This framework is commonly applied in applications such as machine translation, image captioning, and image segmentation. In some advanced models, an attention mechanism is incorporated to allow the decoder to focus on relevant parts of the input, improving the model's ability to handle complex and long-range dependencies.

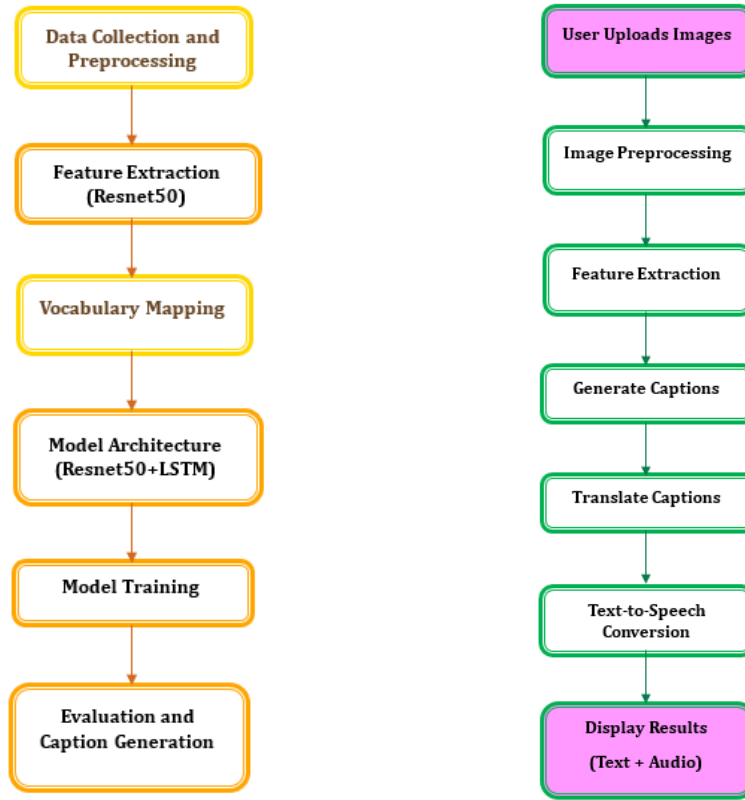


Figure 3.6: Workflow diagram of Training Phase and Inference Phase

3.5.6 Multilingual Integration

1. Using Artificial Tokens

In multilingual captioning systems, artificial tokens are introduced at the beginning of captions to specify the target language. These tokens help the model understand and generate captions in different languages. Each language is associated with a unique token that acts as a language indicator. These tokens are appended to the start of each caption to guide the translation or generation process.

Example Tokens:

- [EN] for English
- [FR] for French
- [ES] for Spanish
- [DE] for German

2. Translation Pipeline

Once a caption is generated in a source language (e.g., English), it can be translated into other languages using translation pipelines. This process involves either using pre-trained models that can generate captions in multiple languages or leveraging external translation APIs, such as Google Translate, to convert the caption into the desired language.

- **Training Data Approach:** A multilingual captioning model can be trained on

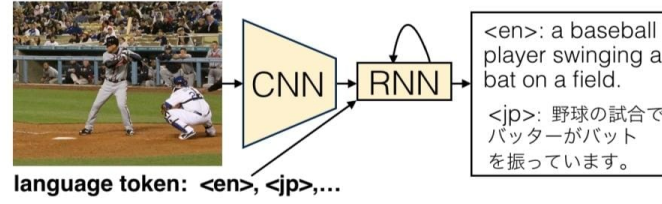


Figure 3.7: Artificial Tokens for Multilingual Captions[6]

data from multiple languages, allowing it to directly generate captions in the target language.

- **API-based Translation:** In cases where multilingual training data is not available, an API like Google Translate or DeepL can be used to translate the generated captions into other languages.

Language	Token	Sample Caption
English	[EN]	A man riding a bicycle.
French	[FR]	Un homme fait du vélo.
Spanish	[ES]	Un hombre montando una bicicleta.
German	[DE]	Ein Mann fährt Fahrrad.

Table 3.2: Language Specific Tokens

3.5.7 Training and Evaluation

1. Training Configuration To train the multilingual captioning model, the following configuration is used:

- **Optimizer:**

Optimizers are algorithms or methods used to adjust the weights of a neural network in order to minimize the loss function and improve the model's performance. They control how the model learns and converges to the optimal parameters.

Adam (Adaptive Moment Estimation) combines the advantages of both Adam and RMSProp optimizers. It computes adaptive learning rates for each parameter by considering both the first-order momentum and second-order velocity of the gradients.

- **Loss Function:**

A loss function is a measure of how well the model's predictions match the

actual outcomes (ground truth). The purpose of the loss function is to quantify the difference between the predicted output and the true value.

Categorical Cross-Entropy is a loss function used in multi-class classification, measuring the difference between the true labels and predicted probabilities. It penalizes the model more when its predictions deviate from the actual class, helping it improve accuracy during training.

- **Batch Size:**

The batch size is the number of training examples used to compute each gradient update during training. It impacts the model's learning process and computational efficiency.

A batch size of 32 is a common choice as it strikes a balance between memory efficiency and computation time. Larger batch sizes require more memory, but smaller batch sizes can increase the variance of the gradients.

- **Epochs:**

Epochs refer to the number of times the entire dataset passes through the model during training. Each epoch consists of multiple iterations, where each iteration processes one batch of data.

2.Evaluation Metrics

To assess the model's performance in generating accurate multilingual captions, the following evaluation metrics are used:

- **BLEU Score (Bilingual Evaluation Understudy):**

BLEU is a widely-used metric that evaluates how closely the generated captions match the reference captions. It calculates the overlap of n-grams (sequences of n words) between the generated and ground truth captions. Higher BLEU scores indicate better alignment between the generated and reference captions, making it an effective measure for evaluating the quality of machine-generated text.

- **Qualitative Analysis:**

This involves human evaluation where the generated captions are compared to the reference captions. Assessors judge the relevance, fluency, and accuracy of the generated captions, ensuring they align with the intended meaning and context. This metric is crucial for capturing subtleties that automated metrics might miss, especially in multilingual settings.

3.5.8 Deployment

Deployment involves integrating the trained model into a web application built with Flask, enabling users to upload images and generate captions in multiple languages.

Additionally, the generated captions are converted into audio using Google Text-to-Speech (gTTS) for enhanced accessibility.

1. Web Application

Flask is used to build the web application as it is lightweight, easy to set up, and highly extensible. It serves as the backend for the application, handling requests and rendering responses.

Features:

- **Image Upload:** Users can upload an image through a simple interface. The application will process the image and generate captions in the desired language(s).
- **Caption Generation in Multiple Languages:** After processing the uploaded image, the model generates captions in multiple languages. These captions are then displayed on the webpage for the user to view.
- **Audio Description Playback:** The generated caption, which could be in any supported language, is converted to audio and played back to the user. This provides accessibility for visually impaired users, ensuring they can also benefit from the generated captions.

2. Audio Description

Google Text-to-Speech (gTTS) is used to convert the generated multilingual captions into speech. It is an efficient tool that allows the conversion of text into natural-sounding audio.

Process

- **Input:** The text of the generated multilingual caption is passed as input to the gTTS tool.
- **Output:** The tool processes the text and generates an audio file in the .mp3 format. This audio file is then made available for playback on the web application, allowing the user to listen to the description in the chosen language.

Chapter 4

RESULTS and DISCUSSION

The Results and Discussion section highlights the evaluation of the model’s performance in generating multilingual captions and the deployment of its functionalities through a Flask-based web interface. Quantitative metrics, qualitative analysis, and user-facing implementations are discussed in detail.

4.1 Result

4.1.1 Model Performance

The model demonstrated strong capabilities in generating captions in multiple languages, including English, French, and Spanish. BLEU scores, a standard metric for evaluating the similarity between generated and reference captions, showed consistent improvement across training epochs. This indicates that the model effectively learned semantic and contextual relationships in the data.

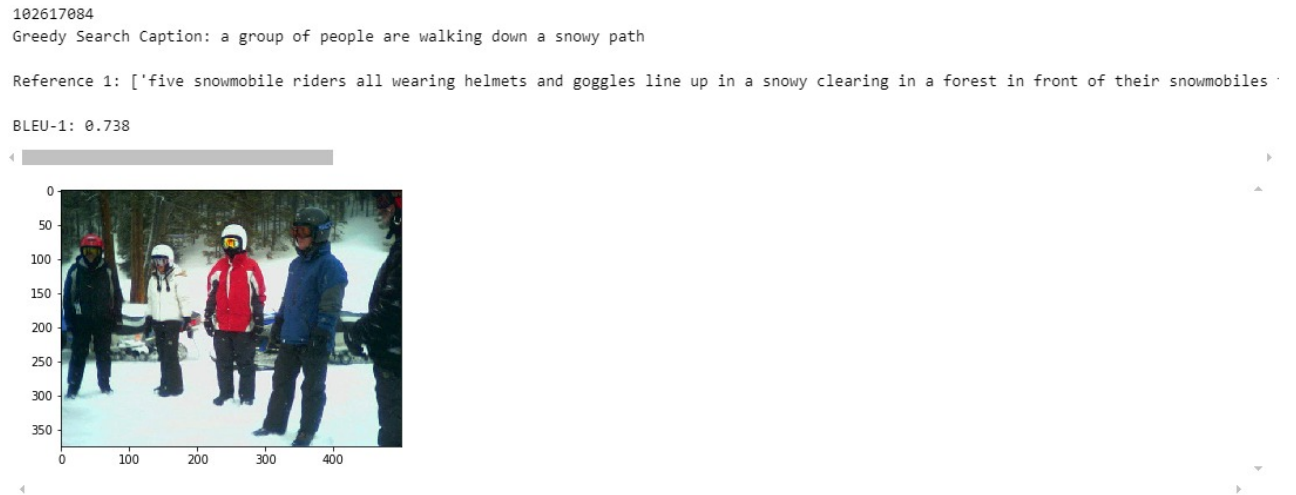


Figure 4.1: BLEU Score Evaluation

4.1.2 Multilingual Capability

The use of artificial tokens ([EN], [FR], [ES]) for specifying target languages enabled the model to handle multilingual captioning with high accuracy. This framework allowed seamless generation of captions in different languages while preserving the contextual and linguistic nuances of each language. Generated captions were evaluated qualitatively and found to align closely with reference captions.

4.1.3 Flask-Based Interface

A user-friendly web interface was developed using Flask to facilitate interaction with the model. The interface allows users to upload images, select a target language for caption generation, and view the results in real time.

Key features of the interface include:

- 1. Image Upload:** Users can upload images for caption generation.
- 2. Language Selection:** A dropdown menu enables users to choose the desired language for captions.
- 3. Generated Caption Display:** Captions are displayed on the interface alongside the uploaded image.
- 4. Audio Playback:** An integrated audio player allows users to listen to captions, generated using Google Text-to-Speech (gTTS).

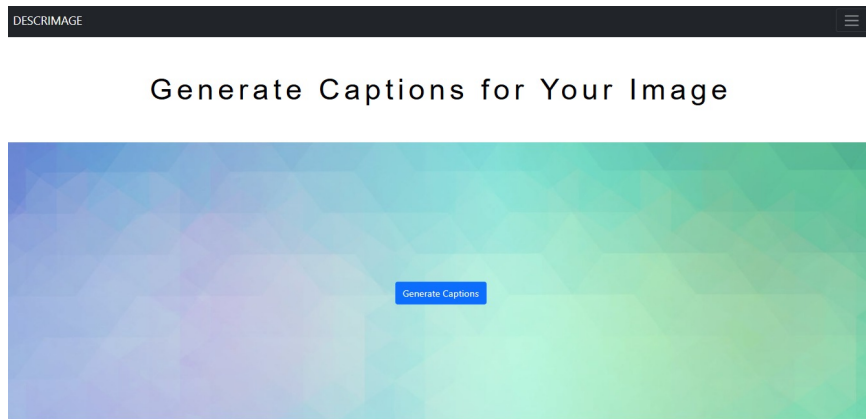


Figure 4.2: Home page

4.1.4 Audio Integration

Google Text-to-Speech (gTTS) was utilized to convert the generated multilingual captions into audio files. The audio descriptions were clear and natural, aligned with the text captions, and effectively broadened the usability of the system, particularly for visually impaired users.

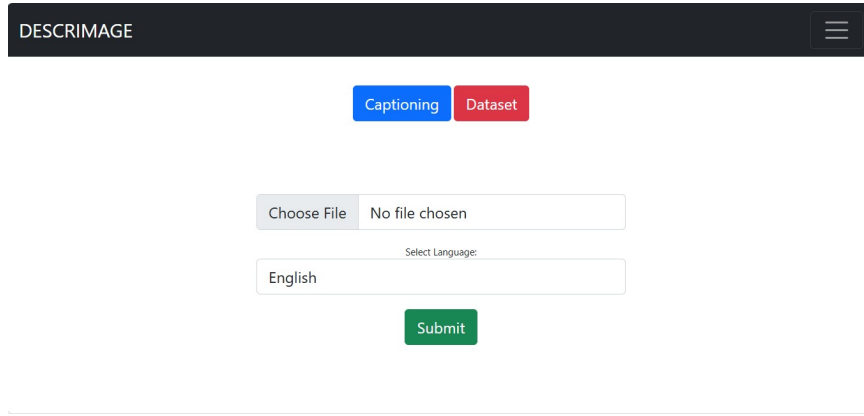


Figure 4.3: Image Upload page

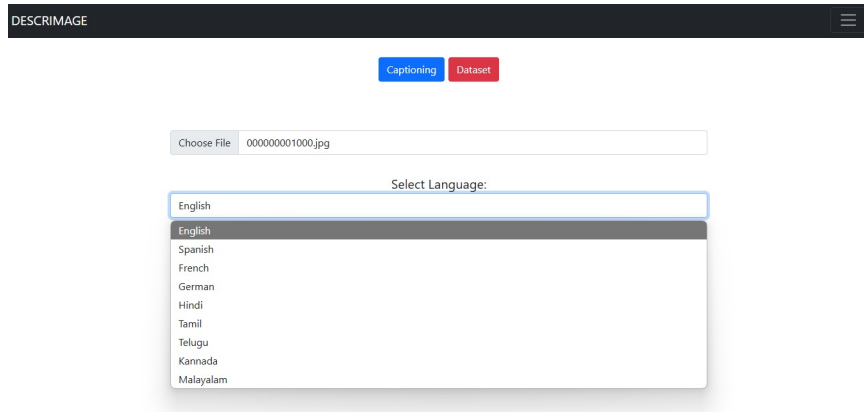


Figure 4.4: Language Selection

4.2 Discussion

The integration of a Flask-based interface significantly enhances the practicality and usability of the system, transforming it from a technical prototype into an accessible and interactive application. This interface allows users to effortlessly upload images, generate multilingual captions, and access audio descriptions, providing a seamless user experience. By simplifying the interaction with complex underlying models, the Flask-based platform bridges the gap between advanced AI technology and everyday usability, making the system approachable for non-technical users.

The consistent improvement in BLEU scores throughout training highlights the model's ability to generalize well across diverse data. This trend reflects the system's capacity to produce accurate and contextually relevant captions, even for unseen images. The BLEU scores are further complemented by qualitative assessments, which confirm the generated captions' fluency and relevance when compared to human-written references. This combination of metrics establishes the reliability of the multilingual captioning framework and its potential for real-world use.

The inclusion of multilingual audio descriptions, powered by Google Text-to-Speech (gTTS), broadens the system's accessibility, particularly for visually impaired

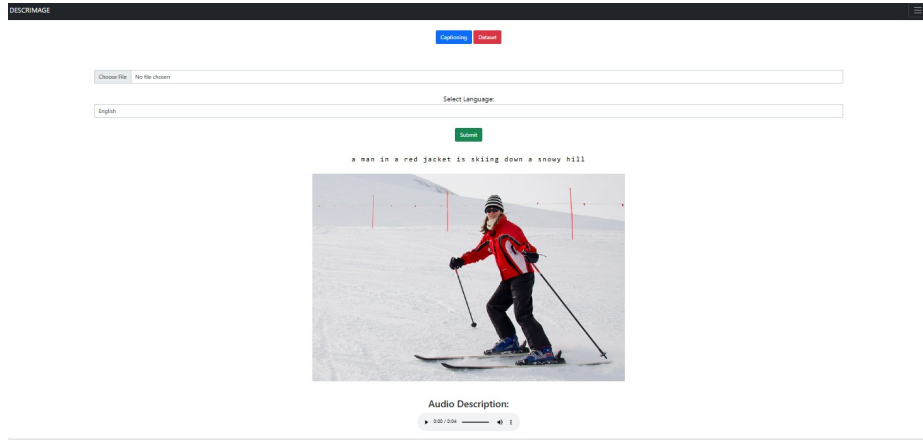


Figure 4.5: Caption Display



Figure 4.6: Audio description

users. By providing high-quality audio descriptions in multiple languages, the system demonstrates its applicability in enhancing accessibility and inclusivity. The synergy between multilingual captioning and audio integration ensures that the system caters to diverse user needs, bridging linguistic and accessibility barriers.

The deployment of a fully functional interface and multilingual capabilities underscores the system’s potential for real-world applications. For example, it can serve as a tool for accessibility by enabling visually impaired individuals to interpret images through audio descriptions. Additionally, the system can be utilized for multilingual content creation, allowing content creators to produce captions in various languages without requiring manual translation. The model’s ability to integrate advanced deep learning techniques with practical deployment frameworks exemplifies its feasibility for impactful applications.

This work highlights the importance of combining robust AI methodologies with accessible interfaces to create practical solutions. While the current system demonstrates significant potential, future improvements could focus on expanding the range of supported languages, optimizing processing speed for real-time applications, and integrating additional features like video captioning. These enhancements would further extend the system’s applicability and impact, ensuring it remains a valuable tool for accessibility and multilingual communication.

Chapter 5

CONCLUSIONS

In conclusion, this project integrates deep learning, multilingual capabilities, and accessibility features to generate image captions and audio descriptions. Using Flask, ResNet50 for feature extraction, LSTM for text generation, and gTTS for text-to-speech, it helps visually impaired users understand images through text and audio. Multilingual support enhances inclusivity, making it accessible globally. Beyond accessibility, the system automates caption generation for education, industry, and research, improving workflows in e-commerce and content creation. Future advancements like real-time video captioning and voice interactions can further enhance its usability, establishing it as a scalable and versatile tool for diverse applications.

5.1 Impacts

This project significantly enhances accessibility for visually impaired individuals, allowing them to engage with visual media through audio descriptions in multiple languages. It supports educational initiatives by providing automated visual aids and captions, and its industrial applications include e-commerce and content creation. Additionally, it demonstrates the power of AI in solving real-world challenges, promoting multilingualism, and fostering cultural inclusivity, all while reducing the resource burden of manual processes.

5.2 Future Advancements

The project can be enhanced by adopting state-of-the-art architectures like transformers for improved captioning accuracy and expanding the multilingual range to include regional dialects. Real-time image and video processing, automatic language detection, and mobile or cloud-based deployment can make the system more scalable and accessible. These advancements could transform the application into a comprehensive tool for global accessibility and communication.

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